Neonatal Tetanus

Contents

Neonatal Tetanus

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Learning objectives

- Definition
- Risk factors / aetiology of neonatal tetanus
- Pathophysiology
- Clinical features and complications

- Differential diagnosis
- Investigations
- Treatment
- Preventive measures

Introduction

Tetanus is an **acute, potentially lethal disease of the nervous system** caused by a toxin (**tetanospasmin**) produced by *Clostridium tetani*.

In neonates, initial symptoms appear **within 3 to 4 days of birth**.

Tetanus is characterised by an acute onset of **hypertonia**, **painful muscular contractions** (usually of the muscles of the jaw and neck), and **generalized muscle spasms** without other apparent medical causes.

Despite widespread immunization of infants and children since the 1940s, tetanus still occurs in the United States.

The organism is found **worldwide in soil, in the inanimate environment, in animal faeces and occasionally human faeces**.

Types of tetanus

1. Generalised tetanus

Manifests within **3-14 days of birth** as progressive difficulty in feeding, associated with hunger and crying, paralysis or diminished movement, stiffness, rigidity to touch, and spasms with or without opisthotonos.

Approximately **50-75%** present with **trismus ("lock jaw")** – inability to open the mouth secondary to masseter muscle spasm.

Nuchal rigidity and **dysphagia** are also early complaints, causing **risus sardonicus**, the scornful smile of tetanus, resulting from facial muscle involvement.

As the disease progresses, patients have generalized muscle spasms in response to stimuli (e.g. noise, touch). Tonic contractions cause **opisthotonos** – flexion and adduction of the arms, clenching of the fists, and extension of the lower extremities.

During these episodes, patients have an **intact sensorium** and feel **severe pain**.

The spasms can cause **fractures, tendon ruptures, and acute respiratory failure**.

2. Localised tetanus

Results in painful spasms of the muscles adjacent to the wound site, and may proceed to generalized tetanus.

The muscular rigidity is caused by a dysfunction in the interneurons that inhibit the alpha motor neurons of the affected muscles. **No further CNS involvement** occurs in this form, and **mortality is very low**.

3. Cephalic tetanus

A rare form of localized tetanus involving the bulbar or facial muscles. Associated with **chronic otitis media**, and usually occurs after head trauma or otitis media.

Characterised by **retracted eyelids, deviated gaze, trismus, risus sardonicus and spastic paralysis of the tongue and pharyngeal musculature**. Patients present with cranial nerve palsies. The infection may be localized or may become generalized.

4. Neonatal tetanus

Also called **tetanus neonatorum**. A major cause of infant mortality in underdeveloped countries, but rare in the U.S.

Infection results from **umbilical cord contamination during unsanitary delivery**, coupled with a **lack of maternal immunization**.

At the end of the first week of life, infected infants become **irritable, feed poorly, and develop rigidity with spasms**.

> Neonatal tetanus has a **very poor prognosis**. Although tetanus is quite rare, early diagnosis and intervention are life saving. **Prevention is the ultimate management strategy.**

Risk factors for neonatal tetanus

- Inadequate immunization of the mother
- Unsterile cutting or care of the umbilicus
- Application of foreign material on the umbilicus, e.g. animal dung
- Stepping on a nail accounted for **32%** of puncture wounds
- Tetanus has been found in burn victims; in patients receiving IM injections; in persons obtaining tattoos; and in persons with frostbite, dental infections (e.g. periodontal abscesses), penetrating eye injuries, and umbilical stump infections
- Other risk factors: diabetes, chronic wounds (chronic wounds, abscesses or gangrene), parenteral drug abuse, and recent surgery

Pathophysiology

The organism

Clostridium tetani is an **obligate anaerobic, motile, gram-positive bacillus**. It is **non-encapsulated** and forms **spores resistant to heat, desiccation and disinfectants**.

Since the colourless spores are located at one end of the bacillus, they cause the organism to resemble a **turkey leg** (drumstick).

They are found in soil, house dust, animal intestines and human faeces. **Spores can persist in normal tissue for months to years.**

Germination

To germinate, the spores require specific **anaerobic conditions**, such as wounds with **low oxidation-reduction potential** – e.g. dead or devitalized tissue, foreign body, active infection. Under these conditions, upon germination, they may release their toxin.

Infection by *C. tetani* results in a **benign appearance at the portal of entry**, because of the inability of the organism to evoke an inflammatory response, unless coinfection with other organisms develops.

The two toxins

Tetanolysin – a haemolysin with **no recognized pathologic activity**.

Tetanospasmin – responsible for the clinical manifestations of tetanus.

- By weight it is **one of the most potent toxins known**, with an estimated **minimum lethal dose of 2.5 ng/kg** body weight
- Synthesized as a **150-kd protein**, consisting of a **100-kd heavy chain** and a **50-kd light chain** joined by a **disulphide bond**
- The **heavy chain** mediates binding of tetanospasmin to the **presynaptic motor neuron**

Mechanism of action

After the **light chain** enters the motor neuron, it travels by **retrograde axonal transport** from the contaminated site to the spinal cord in **2-14 days**.

When the toxin reaches the spinal cord, it enters **central inhibitory neurons**. The light chain **cleaves the protein synaptobrevin**, which is integral to the binding of neurotransmitter-containing vesicles to the cell membrane.

GABA- and glycine-containing vesicles are not released, leading to **loss of inhibitory action** on motor and autonomic neurons.

The result is **autonomic hyperactivity** as well as **uncontrolled muscle contraction (spasms)** in response to normal stimuli such as noises or lights.

> Once the toxin becomes **fixed to neurons, it cannot be neutralized with antitoxin**. Recovery of nerve function requires **sprouting of new nerve terminals and formation of new synapses**.

Clinical features

- Poor sucking, difficulty in swallowing – then followed by
- Generalised muscle rigidity and painful spasm (e.g. opisthotonos position, stiffness of the jaw, laryngospasm)
- Restlessness, irritability
- Difficulty in breathing, fast breathing, apnoea, cyanosis
- Weak cry
- Sardonic smile (risus sardonicus)

> **NOTE:** Spasms can be set off by disturbances such as **noises, light, and touch**.

Complications

- Aspiration pneumonia
- Pneumothorax
- Seizures
- Spinal fractures
- Pulmonary embolism

- Gastric ulceration
- Mediastinal emphysema

Differential diagnosis

- Acute encephalitis
- Epileptic seizures
- Parapharyngeal abscess
- Retropharyngeal abscess
- Rabies
- Meningitis / encephalitis

Conditions producing trismus: alveolar abscess, strychnine poisoning, dystonic drug reactions, hypocalcaemic tetany.

> **Marked increased tone in central muscles with superimposed generalized spasms and relative sparing of the hands and feet – suggests tetanus.**

Investigations

The diagnosis is **basically clinical**, with proper history taking and examination. **There are no specific confirmatory tests.**

Others:

- **Peripheral leukocytosis** – results from secondary infection of the wound
- **CSF** – although the intense muscle contractions may raise intracranial pressure
- Electroencephalogram
- Blood culture
- Random blood glucose

General management measures

- **Supportive measures** – IV fluids, oxygen, a quiet dark room
- **Antibiotics** – to eradicate the source of tetanus
- **Antitoxin (TIG or TAT)** – to neutralize circulating toxin
- **Control of muscle spasms** – diazepam
- **Respiratory care** – gentle suctioning, tracheostomy
- **Management of complications**

Supportive measures (non-pharmacological)

- Control light and noise in the room, to avoid provoking spasms
- **Rigorously cleanse the umbilical stump** to stop the production of toxin at the site of infection

Antibiotics

- **Amoxicillin-clavulanate** (po) via nasogastric tube, 20-30 mg/kg/day divided 8 hourly for 7 days, **AND**
- **Metronidazole** (po) 7.5 mg/kg for postnatal age <= 7 days; weighing 1200-2000 g: 7.5 mg/kg/day every 24 hours; > 2000 g: 15 mg/kg/day in divided doses every 12 hours; > 2000 g: 30 mg/kg/day in divided doses every 12 hours, for 7 days, **OR**
- **Ceftriaxone** (IV) 2 g (50 mg/kg in paediatric patients older than 1 month) 12 hourly for 7 days

- **Cefotaxime** (IV) 2 g (50 mg/kg) in paediatric patients older than 1 month, 6 hourly for 5 days

Antitoxin

- **Tetanus immunoglobulin (TIG / human tetanus immunoglobulin)** – given **intramuscularly** in a single dose of **250-500 units**
- If human serum immunoglobulin is unavailable, **tetanus antitoxin (TAT) 1500 IU** single dose IM should be given, assuming sensitivity reactions to horse serum are negative
- In addition, a **single dose of the DT vaccine** is recommended

Control of muscle spasms

- **Diazepam** 0.1-0.2 mg/kg every 3-6 hours intravenously, subsequently titrated to control the tetanic spasms
- **Chlorpromazine** 1-5 mg/kg/dose every 8 hours
- **Phenobarbitone** – loading dose 20 mg/kg, then 2.5 mg/kg/dose every 12 hours, increased to a maximum of 5 mg/kg/dose every 12 hours

Rotating schedule (hours):

- Diazepam – 0, 3, 9, 15, 21, 24
- Chlorpromazine – 3, 9, 15, 21
- Phenobarbitone – 0, 6 and 21

Management of complications

Specific therapy for autonomic system complications and control of spasms should be initiated.

Magnesium sulphate can be used alone or in combination with benzodiazepines for this purpose.

- Given IV in a loading dose of **5 g (or 75 mg/kg)**
- Followed by continuous infusion at a rate of **2.3 g/h** until spasm control is achieved
- The **patellar reflex should be monitored**; **areflexia** (absence of the patellar reflex) occurs at the upper end of the therapeutic range (**4 mmol/L**). If areflexia develops, the dosage should be reduced

Prevention

- Immunise the mother and other pregnant women in the same locality as the case with **at least 2 doses of tetanus toxoid**
- Conduct a **supplemental immunization activity** for women of childbearing age in the locality
- Improve routine vaccine coverage and maternal immunization programme activities
- **Educate birth attendants and women of childbearing age** on the need for clean cord cutting and care; increase the number of trained birth attendants
- **TT boosters at 10 years**
- Prompt cleaning of wounds with **hydrogen peroxide**
- **Tetanus is completely preventable by active tetanus immunization**
- Cutting of the umbilical stump with a **sterile instrument**
- Observing simple cleanliness at birth, and hygienic handling and care of the umbilical cord

Prognosis

Mortality is about 90%.

Bad prognostic signs:

1. Onset in the first week of life
2. Interval between lockjaw (trismus) and onset of muscle spasms **less than 48 hours**
3. High fever
4. Tachycardia

Recovery is almost complete.

> **An attack of tetanus does not confer immunity**, so active immunization following recovery is a must.